removed through averaging out all numbers into 10 groups by their tens digit.

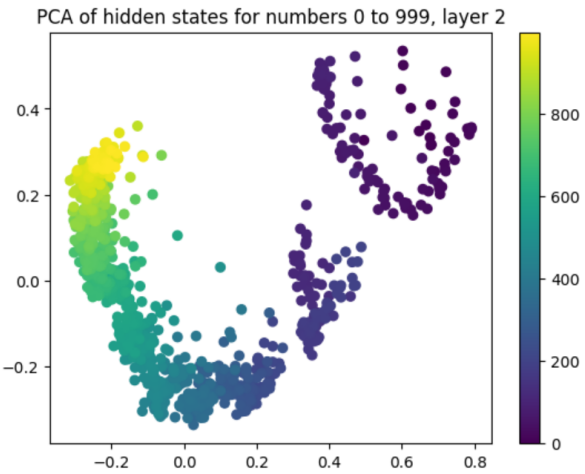

Figure 5: Visualization of the hidden states for natural number tokens (0 to 999) in layer 2 of Llama 3 8B, projected onto their top two principal components.

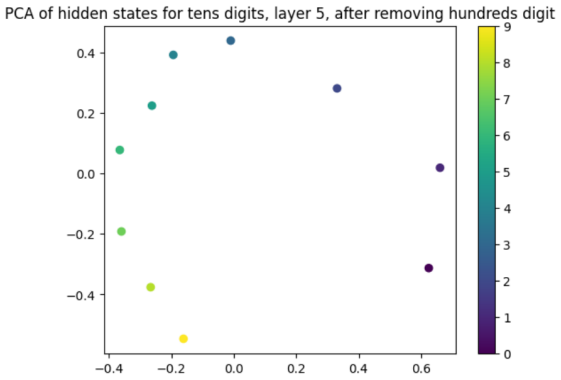

Figure 6: Visualization of the averaged hidden states for natural number tokens (0 to 999), grouped by their tens digit (0—9), in layer 5 of Llama 3 8B, projected onto their top two principal components. For example, numbers like 101 and 406, both having a tens digit of 0, are grouped together.

A.3 Accuracy of circular probes

In the main results we showed the accuracy of the circular digit-wise probes, averaged over layers ≥ 3. Here we will show this choice is justified as can be seen in Figure 7. While there is significant variations between layers, the accuracy is especially low before the contextualization that happens in the first 3 layers.

Another interesting question is which layer's set of digit-wise circular-probes have the highest accuracy in predicting the number, and how accurate is it. The corresponding results can be seen in Table 3. It can be observed that in Mistral 7B, in the best layer, the circular probes achieve perfect accuracy on the validation set. That is, the number can always be recreated perfectly.

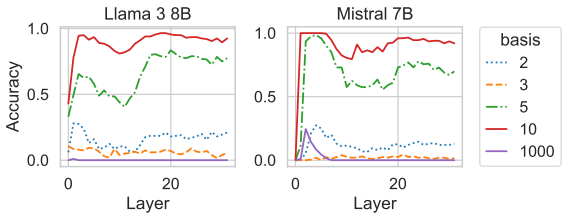

Figure 7: Accuracy of the circular probes in different bases across layers in Llama (left) and Mistral (right).

A.4 Probing representations of numbers in word form

Since numbers can also be represented in word form, i.e. twenty-two for the number 22, we further tested if our digit-wise circular probes extend to these representations, without being explicitly trained on them. Concretely, we evaluated the probes' accuracy for Llama 3 8B on the numbers 'zero' through 'fifty' in word form.

We observe that the accuracy varies depending on the layer used, with a peak of 68.6 accuracy when using the representations at layer 14. This is an encouraging sign that the circular probes generalize beyond the specific setting they were trained on, which further supports our causal results.

B Causal Intervention Details

We provide additional details on the interventions performed in §3.3. In practice, since the two directions of the circular probe are approximately orthogonal, we project the hidden representation onto each direction, subtract these components to remove the original digit representation, and then add the components back with their directions reversed to modify the digit. We also scaled the projection by a fixed constant ($a = 19$), assuming that if the model has multiple representations for numbers, scaling the representation will make the model place more weight upon it. The exact constant was chosen through binary search, in order to select the largest scaling factor such that the model still predicts a number, as it was observed that with a very high scaling factor the model starts predicting non-numeric tokens.

Basis	2	3	4	5	6	7	8	9	10	11	12	13	14	1000	2000
Llama 3 8B	0.24	0.10	0.25	<u>0.84</u>	0.08	0.10	0.10	0.11	**0.96**	0.12	0.09	0.10	0.11	0.00	0.02
Mistral 7B	0.28	0.04	0.22	<u>0.98</u>	0.04	0.08	0.12	0.23	**1.00**	0.29	0.08	0.18	0.10	0.14	0.03

Table 3: Accuracy of the digit-wise circular probes for different bases in predicting all digits correctly, taking the layer with the highest accuracy.